Next, note that

$$f(z) = \frac{-iz}{z^2} + E(z)$$

where $E(z)$ is bounded as $z \to 0$, while on γ_ϵ^+ we have $z = \epsilon e^{i\theta}$ and $dz = i\epsilon e^{i\theta} d\theta$. Thus

$$\int_{\gamma_\epsilon^+} \frac{1 - e^{iz}}{z^2}\, dz \to \int_\pi^0 (-ii)\, d\theta = -\pi \qquad \text{as } \epsilon \to 0.$$

Taking real parts then yields

$$\int_{-\infty}^{\infty} \frac{1 - \cos x}{x^2}\, dx = \pi.$$

Since the integrand is even, the desired formula is proved.

4 Cauchy's integral formulas

Representation formulas, and in particular integral representation formulas, play an important role in mathematics, since they allow us to recover a function on a large set from its behavior on a smaller set. For example, we saw in Book I that a solution of the steady-state heat equation in the disc was completely determined by its boundary values on the circle via a convolution with the Poisson kernel

$$u(r, \theta) = \frac{1}{2\pi} \int_0^{2\pi} P_r(\theta - \varphi) u(1, \varphi)\, d\varphi. \tag{7}$$

In the case of holomorphic functions, the situation is analogous, which is not surprising since the real and imaginary parts of a holomorphic function are harmonic.[3] Here, we will prove an integral representation formula in a manner that is independent of the theory of harmonic functions. In fact, it is also possible to recover the Poisson integral formula (7) as a consequence of the next theorem (see Exercises 11 and 12).

Theorem 4.1 *Suppose f is holomorphic in an open set that contains the closure of a disc D. If C denotes the boundary circle of this disc with the positive orientation, then*

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta \qquad \textit{for any point } z \in D.$$

[3] This fact is an immediate consequence of the Cauchy-Riemann equations. We refer the reader to Exercise 11 in Chapter 1.

Proof. Fix $z \in D$ and consider the "keyhole" $\Gamma_{\delta,\epsilon}$ which omits the point z as shown in Figure 10.

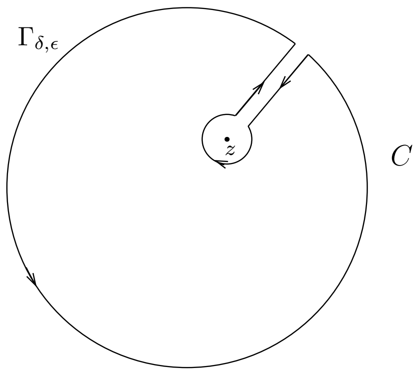

Figure 10. The keyhole $\Gamma_{\delta,\epsilon}$

Here δ is the width of the corridor, and ϵ the radius of the small circle centered at z. Since the function $F(\zeta) = f(\zeta)/(\zeta - z)$ is holomorphic away from the point $\zeta = z$, we have

$$\int_{\Gamma_{\delta,\epsilon}} F(\zeta)\, d\zeta = 0$$

by Cauchy's theorem for the chosen toy contour. Now we make the corridor narrower by letting δ tend to 0, and use the continuity of F to see that in the limit, the integrals over the two sides of the corridor cancel out. The remaining part consists of two curves, the large boundary circle C with the positive orientation, and a small circle C_ϵ centered at z of radius ϵ and oriented negatively, that is, clockwise. To see what happens to the integral over the small circle we write

$$F(\zeta) = \frac{f(\zeta) - f(z)}{\zeta - z} + \frac{f(z)}{\zeta - z} \tag{8}$$

and note that since f is holomorphic the first term on the right-hand side of (8) is bounded so that its integral over C_ϵ goes to 0 as $\epsilon \to 0$. To conclude the proof, it suffices to observe that

$$\begin{aligned}\int_{C_\epsilon} \frac{f(z)}{\zeta - z}\, d\zeta &= f(z) \int_{C_\epsilon} \frac{d\zeta}{\zeta - z} \\ &= -f(z) \int_0^{2\pi} \frac{\epsilon i e^{-it}}{\epsilon e^{-it}}\, dt \\ &= -f(z) 2\pi i\,,\end{aligned}$$

so that in the limit we find

$$0 = \int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta - 2\pi i f(z)\,,$$

as was to be shown.

Remarks. Our earlier discussion of toy contours provides simple extensions of the Cauchy integral formula; for instance, if f is holomorphic in an open set that contains a (positively oriented) rectangle R and its interior, then

$$f(z) = \frac{1}{2\pi i} \int_R \frac{f(\zeta)}{\zeta - z}\, d\zeta\,,$$

whenever z belongs to the interior of R. To establish this result, it suffices to repeat the proof of Theorem 4.1 replacing the "circular" keyhole by a "rectangular" keyhole.

It should also be noted that the above integral vanishes when z is outside R, since in this case $F(\zeta) = f(\zeta)/(\zeta - z)$ is holomorphic inside R. Of course, a similar result also holds for the circle or any other toy contour.

As a corollary to the Cauchy integral formula, we arrive at a second remarkable fact about holomorphic functions, namely their regularity. We also obtain further integral formulas expressing the derivatives of f inside the disc in terms of the values of f on the boundary.

Corollary 4.2 *If f is holomorphic in an open set Ω, then f has infinitely many complex derivatives in Ω. Moreover, if $C \subset \Omega$ is a circle whose interior is also contained in Ω, then*

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}}\, d\zeta$$

for all z in the interior of C.

We recall that, as in the above theorem, we take the circle C to have positive orientation.

Proof. The proof is by induction on n, the case $n = 0$ being simply the Cauchy integral formula. Suppose that f has up to $n-1$ complex derivatives and that

$$f^{(n-1)}(z) = \frac{(n-1)!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^n}\, d\zeta.$$

Now for h small, the difference quotient for $f^{(n-1)}$ takes the form

$$\frac{f^{(n-1)}(z+h) - f^{(n-1)}(z)}{h} = \tag{9}$$

$$\frac{(n-1)!}{2\pi i} \int_C f(\zeta) \frac{1}{h} \left[\frac{1}{(\zeta - z - h)^n} - \frac{1}{(\zeta - z)^n} \right] d\zeta.$$

We now recall that

$$A^n - B^n = (A - B)[A^{n-1} + A^{n-2}B + \cdots + AB^{n-2} + B^{n-1}].$$

With $A = 1/(\zeta - z - h)$ and $B = 1/(\zeta - z)$, we see that the term in brackets in equation (9) is equal to

$$\frac{h}{(\zeta - z - h)(\zeta - z)} [A^{n-1} + A^{n-2}B + \cdots + AB^{n-2} + B^{n-1}].$$

But observe that if h is small, then $z + h$ and z stay at a finite distance from the boundary circle C, so in the limit as h tends to 0, we find that the quotient converges to

$$\frac{(n-1)!}{2\pi i} \int_C f(\zeta) \left[\frac{1}{(\zeta - z)^2} \right] \left[\frac{n}{(\zeta - z)^{n-1}} \right] d\zeta = \frac{n!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}}\, d\zeta\,,$$

which completes the induction argument and proves the theorem.

From now on, we call the formulas of Theorem 4.1 and Corollary 4.2 the **Cauchy integral formulas**.

Corollary 4.3 (Cauchy inequalities) *If f is holomorphic in an open set that contains the closure of a disc D centered at z_0 and of radius R, then*

$$|f^{(n)}(z_0)| \le \frac{n! \|f\|_C}{R^n}\,,$$

where $\|f\|_C = \sup_{z \in C} |f(z)|$ denotes the supremum of $|f|$ on the boundary circle C.

Proof. Applying the Cauchy integral formula for $f^{(n)}(z_0)$, we obtain

$$\begin{aligned} |f^{(n)}(z_0)| &= \left| \frac{n!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\, d\zeta \right| \\ &= \frac{n!}{2\pi} \left| \int_0^{2\pi} \frac{f(z_0 + Re^{i\theta})}{(Re^{i\theta})^{n+1}} Rie^{i\theta}\, d\theta \right| \\ &\leq \frac{n!}{2\pi} \frac{\|f\|_C}{R^n} 2\pi. \end{aligned}$$

Another striking consequence of the Cauchy integral formula is its connection with power series. In Chapter 1, we proved that a power series is holomorphic in the interior of its disc of convergence, and promised a proof of a converse, which is the content of the next theorem.

Theorem 4.4 *Suppose f is holomorphic in an open set Ω. If D is a disc centered at z_0 and whose closure is contained in Ω, then f has a power series expansion at z_0*

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

for all $z \in D$, and the coefficients are given by

$$a_n = \frac{f^{(n)}(z_0)}{n!} \qquad \text{for all } n \geq 0.$$

Proof. Fix $z \in D$. By the Cauchy integral formula, we have

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta\,, \tag{10}$$

where C denotes the boundary of the disc and $z \in D$. The idea is to write

$$\frac{1}{\zeta - z} = \frac{1}{\zeta - z_0 - (z - z_0)} = \frac{1}{\zeta - z_0} \frac{1}{1 - \left(\frac{z - z_0}{\zeta - z_0}\right)}, \tag{11}$$

and use the geometric series expansion. Since $\zeta \in C$ and $z \in D$ is fixed, there exists $0 < r < 1$ such that

$$\left| \frac{z - z_0}{\zeta - z_0} \right| < r,$$

therefore

$$\frac{1}{1-\left(\frac{z-z_0}{\zeta-z_0}\right)}=\sum_{n=0}^{\infty}\left(\frac{z-z_0}{\zeta-z_0}\right)^n, \tag{12}$$

where the series converges uniformly for $\zeta \in C$. This allows us to interchange the infinite sum with the integral when we combine (10), (11), and (12), thereby obtaining

$$f(z)=\sum_{n=0}^{\infty}\left(\frac{1}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta-z_0)^{n+1}}\,d\zeta\right)\cdot(z-z_0)^n.$$

This proves the power series expansion; further the use of the Cauchy integral formulas for the derivatives (or simply differentiation of the series) proves the formula for a_n.

Observe that since power series define indefinitely (complex) differentiable functions, the theorem gives another proof that a holomorphic function is automatically indefinitely differentiable.

Another important observation is that the power series expansion of f centered at z_0 converges in any disc, no matter how large, as long as its closure is contained in Ω. In particular, if f is entire (that is, holomorphic on all of $\mathbb{C}$), the theorem implies that f has a power series expansion around 0, say $f(z)=\sum_{n=0}^{\infty}a_n z^n$, that converges in all of $\mathbb{C}$.

Corollary 4.5 (Liouville's theorem) *If f is entire and bounded, then f is constant.*

Proof. It suffices to prove that $f'=0$, since $\mathbb{C}$ is connected, and we may then apply Corollary 3.4 in Chapter 1.

For each $z_0 \in \mathbb{C}$ and all $R>0$, the Cauchy inequalities yield

$$|f'(z_0)| \le \frac{B}{R}$$

where B is a bound for f. Letting $R\to\infty$ gives the desired result.

As an application of our work so far, we can give an elegant proof of the fundamental theorem of algebra.

Corollary 4.6 *Every non-constant polynomial $P(z)=a_n z^n+\cdots+a_0$ with complex coefficients has a root in $\mathbb{C}$.*

Proof. If P has no roots, then $1/P(z)$ is a bounded holomorphic function. To see this, we can of course assume that $a_n \neq 0$, and write

$$\frac{P(z)}{z^n}=a_n+\left(\frac{a_{n-1}}{z}+\cdots+\frac{a_0}{z^n}\right)$$

whenever $z \neq 0$. Since each term in the parentheses goes to 0 as $|z| \to \infty$ we conclude that there exists $R > 0$ so that if $c = |a_n|/2$, then

$$|P(z)| \geq c|z|^n \quad \text{whenever } |z| > R.$$

In particular, P is bounded from below when $|z| > R$. Since P is continuous and has no roots in the disc $|z| \leq R$, it is bounded from below in that disc as well, thereby proving our claim.

By Liouville's theorem we then conclude that $1/P$ is constant. This contradicts our assumption that P is non-constant and proves the corollary.

Corollary 4.7 *Every polynomial $P(z) = a_n z^n + \cdots + a_0$ of degree $n \geq 1$ has precisely n roots in $\mathbb{C}$. If these roots are denoted by $w_1, \ldots, w_n$, then P can be factored as*

$$P(z) = a_n(z - w_1)(z - w_2) \cdots (z - w_n).$$

Proof. By the previous result P has a root, say w_1. Then, writing $z = (z - w_1) + w_1$, inserting this expression for z in P, and using the binomial formula we get

$$P(z) = b_n(z - w_1)^n + \cdots + b_1(z - w_1) + b_0,$$

where $b_0, \ldots, b_{n-1}$ are new coefficients, and $b_n = a_n$. Since $P(w_1) = 0$, we find that $b_0 = 0$, therefore

$$P(z) = (z - w_1)\left[b_n(z - w_1)^{n-1} + \cdots + b_1\right] = (z - w_1)Q(z),$$

where Q is a polynomial of degree $n - 1$. By induction on the degree of the polynomial, we conclude that $P(z)$ has n roots and can be expressed as

$$P(z) = c(z - w_1)(z - w_2) \cdots (z - w_n)$$

for some $c \in \mathbb{C}$. Expanding the right-hand side, we realize that the coefficient of z^n is c and therefore $c = a_n$ as claimed.

Finally, we end this section with a discussion of analytic continuation (the third of the "miracles" we mentioned in the introduction). It states that the "genetic code" of a holomorphic function is determined (that is, the function is fixed) if we know its values on appropriate arbitrarily small subsets. Note that in the theorem below, Ω is assumed connected.

Theorem 4.8 *Suppose f is a holomorphic function in a region Ω that vanishes on a sequence of distinct points with a limit point in Ω. Then f is identically* 0.

In other words, if the zeros of a holomorphic function f in the connected open set Ω accumulate *in* Ω, then $f = 0$.

Proof. Suppose that $z_0 \in \Omega$ is a limit point for the sequence $\{w_k\}_{k=1}^{\infty}$ and that $f(w_k) = 0$. First, we show that f is identically zero in a small disc containing z_0. For that, we choose a disc D centered at z_0 and contained in Ω, and consider the power series expansion of f in that disc

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n.$$

If f is not identically zero, there exists a smallest integer m such that $a_m \neq 0$. But then we can write

$$f(z) = a_m (z - z_0)^m (1 + g(z - z_0)),$$

where $g(z - z_0)$ converges to 0 as $z \to z_0$. Taking $z = w_k \neq z_0$ for a sequence of points converging to z_0, we get a contradiction since $a_m (w_k - z_0)^m \neq 0$ and $1 + g(w_k - z_0) \neq 0$, but $f(w_k) = 0$.

We conclude the proof using the fact that Ω is connected. Let U denote the interior of the set of points where $f(z) = 0$. Then U is open by definition and non-empty by the argument just given. The set U is also closed since if $z_n \in U$ and $z_n \to z$, then $f(z) = 0$ by continuity, and f vanishes in a neighborhood of z by the argument above. Hence $z \in U$. Now if we let V denote the complement of U in Ω, we conclude that U and V are both open, disjoint, and

$$\Omega = U \cup V.$$

Since Ω is connected we conclude that either U or V is empty. (Here we use one of the two equivalent definitions of connectedness discussed in Chapter 1.) Since $z_0 \in U$, we find that $U = \Omega$ and the proof is complete.

An immediate consequence of the theorem is the following.

Corollary 4.9 *Suppose f and g are holomorphic in a region Ω and $f(z) = g(z)$ for all z in some non-empty open subset of Ω (or more generally for z in some sequence of distinct points with limit point in Ω). Then $f(z) = g(z)$ throughout Ω.*

Suppose we are given a pair of functions f and F analytic in regions Ω and Ω', respectively, with $\Omega \subset \Omega'$. If the two functions agree on the smaller set Ω, we say that F is an **analytic continuation** of f into the region Ω'. The corollary then guarantees that there can be only one such analytic continuation, since F is uniquely determined by f.

5 Further applications

We gather in this section various consequences of the results proved so far.

5.1 Morera's theorem

A direct application of what was proved here is a converse of Cauchy's theorem.

Theorem 5.1 *Suppose f is a continuous function in the open disc D such that for any triangle T contained in D*

$$\int_T f(z)\,dz = 0,$$

then f is holomorphic.

Proof. By the proof of Theorem 2.1 the function f has a primitive F in D that satisfies $F' = f$. By the regularity theorem, we know that F is indefinitely (and hence twice) complex differentiable, and therefore f is holomorphic.

5.2 Sequences of holomorphic functions

Theorem 5.2 *If $\{f_n\}_{n=1}^{\infty}$ is a sequence of holomorphic functions that converges uniformly to a function f in every compact subset of Ω, then f is holomorphic in Ω.*

Proof. Let D be any disc whose closure is contained in Ω and T any triangle in that disc. Then, since each f_n is holomorphic, Goursat's theorem implies

$$\int_T f_n(z)\,dz = 0 \qquad \text{for all } n.$$

By assumption $f_n \to f$ uniformly in the closure of D, so f is continuous and

$$\int_T f_n(z)\,dz \to \int_T f(z)\,dz.$$